Temporal and identity predictions shape sensory responses to sound in interaction with attention

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1. Intro: Predicting "what" and "when"

- Prediction of the timing ("when") and identity ("what") of sounds optimizes perception[1]
- Auditory evoked potentials (AEPs), including the P1 and N1 responses, are attenuated for predicted sounds[2,3]

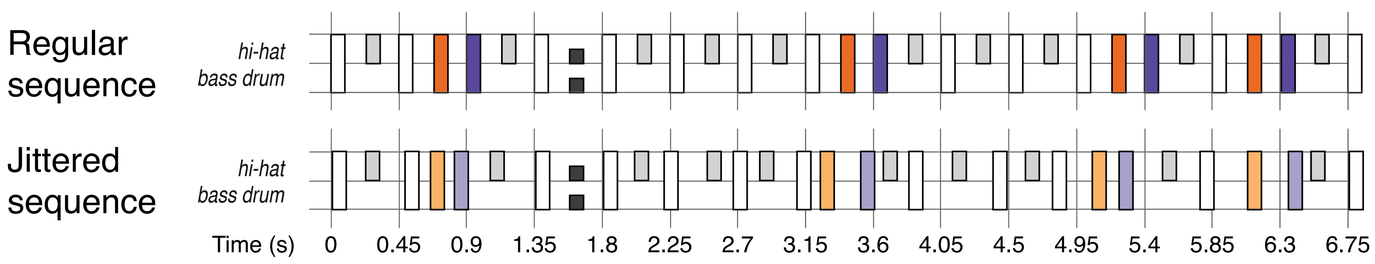
Do temporal and identity predictions interact in affecting early sensory processing, in the form of P1 and N1 responses?

- Temporal predictions may enhance identity predictions[4]
- Temporal and identity predictions may only interact at later stages in processing[3]

What is the influence of task-relevance on how predictions shape perception?

- Attention may enhance the effect of predictions[5]
- Attention may cancel the attenuating effect of predictions[6]

2. Methods: Temporal and identity predictions in rhythmic sounds



Tones of interest

- $T^+ I^+$ (Regular Beat)
- $T^+ I^-$ (Regular Offbeat)
- $T^- I^+$ (Jittered Beat)
- $T^- I^-$ (Jittered Offbeat)

Other

- Beat after hi-hat
- Deviant (See Bouwer et al., 2016[7])
- Offbeat hi-hat

Sound examples available!

Identity predictions

- I^+ Uneven sounds are always bass drum sounds (predictable identity, "Beat")
- I^- Even sounds are bass drum sounds 30% of the time (unpredictable identity, "Offbeat")

Temporal predictions

- T^+ Regular sequences with inter-onset intervals of always 225 ms (predictable timing)
- T^- Jittered sequences with inter-onset intervals between 150 and 300 ms (unpredictable timing)

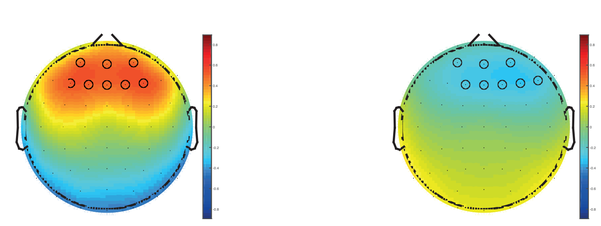
Task-relevance

- A^+ Participants perform a task related to the rhythm (attended)
- A^- Participants watch a muted movie (unattended)

3. Methods: Measuring early sensory ERP responses

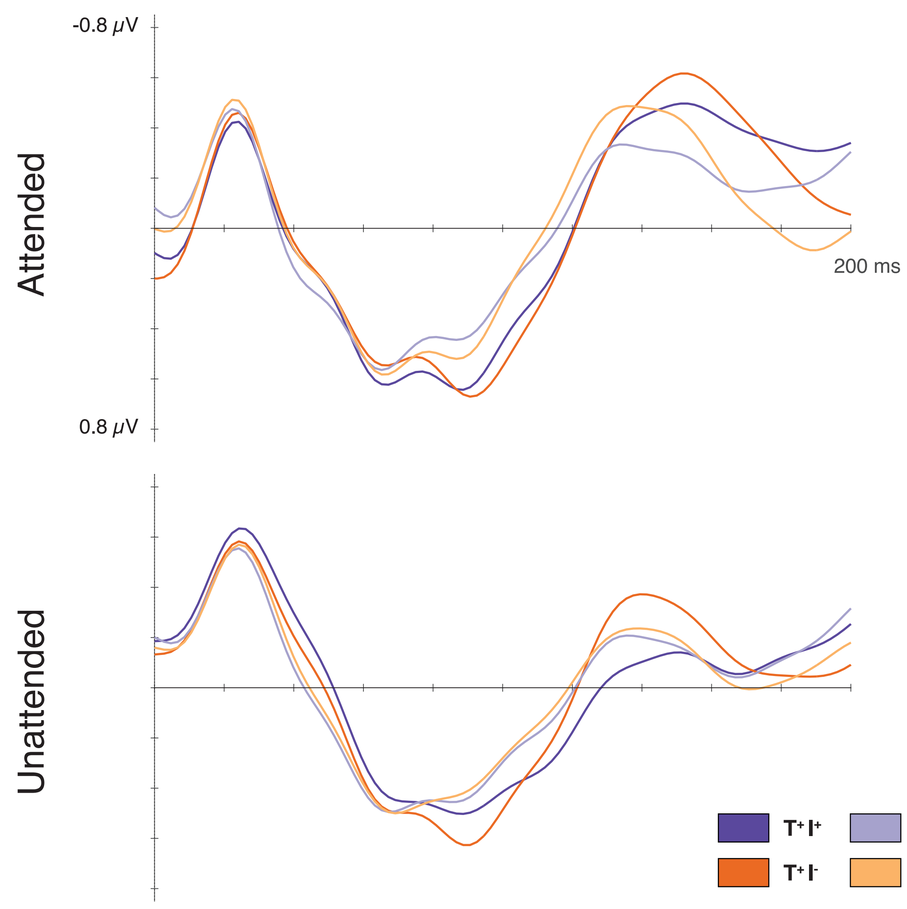
Data acquisition and preprocessing

- 34 participants, 64 channel EEG (Biosemi), ~750 trials per condition
- Downsampled to 512 Hz; re-referenced to average; filtered 1-40 Hz
- Epochs 0 to 200 ms relative to onsets; baseline-corrected with whole epoch (0-200 ms)
- Waveforms averaged over all conditions to define time windows and electrodes of interest
- 2x2x2 ANOVA; mean amplitude from 8 electrodes with maximal responses
- 20 ms (P1) and 40 ms (N1) windows around the peaks

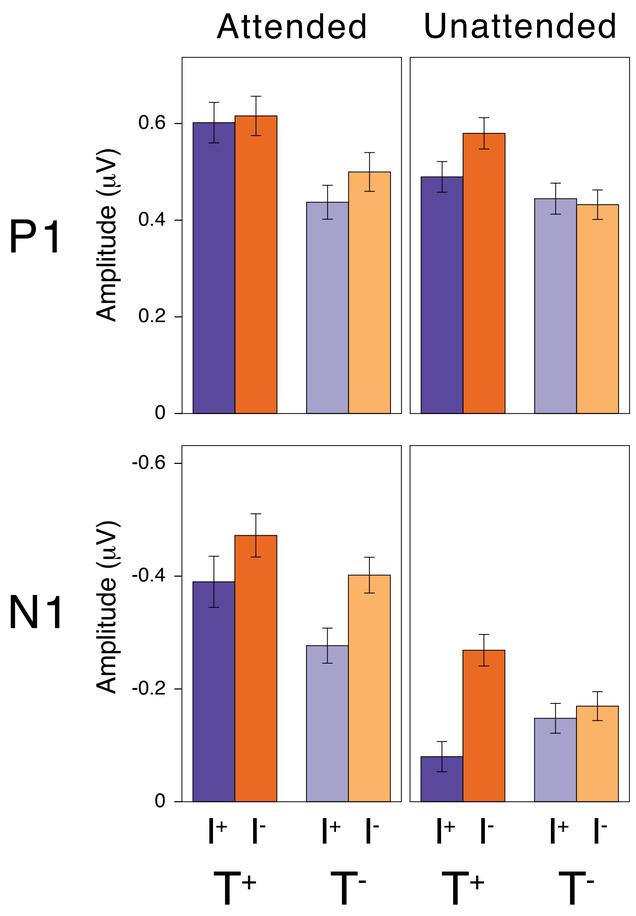


P1 (77-97 ms) N1 (123-163 ms)

4. Results: ERP waveforms



4. Results: Timing, identity, and attention interact



P1

Timing * Identity * Attention (p=0.034)

P1 Attended

Timing (p=0.001); $T^+ > T^-$

P1 Unattended

Timing * Identity (p=0.037)

- $I^- > I^+$, but only for T^+ (p=0.015)
- $T^+ > T^-$, but only for I^- (p<0.001)

N1

Timing * Identity * Attention (p<0.001)

N1 Attended

Timing (p=0.017); $T^+ > T^-$

Identity (p=0.017); $I^- > I^+$

N1 Unattended

Timing * Identity (p<0.001)

- $I^- > I^+$, but only for T^+ (p<0.001)
- $T^+ > T^-$ for I^- (p<0.001)
- $T^- > T^+$ for I^+ (p<0.028)

5. Conclusion

- Identity predictions seem to *attenuate* sensory responses to sound
- Temporal predictions seem to *enhance* sensory responses to sound

Temporal predictions, identity predictions, and attention interact

- Attenuation for *predictable identity* is:
  - Absent with *unpredictable timing* and *unattended* sounds
  - Largest with *predictable timing* and *unattended* sounds
- This may be explained by assuming that:
  - Attention is necessary for identity predictions with unpredictable timing
  - Attention (partly) cancels the attenuation caused by prediction

Food for thought

- Is enhancement of P1 and N1 in the regular condition caused by fluctuations in attention instead of prediction[2]?
- Are temporal predictions in the regular condition beat-based or memory-based[8]?
- How do temporal predictions at a higher level (beat perception) affect our results[7]?

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